# Sea Winds Predictions, Phase 2 Report

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## Executive summary

The Phase 2 task joins three decisions: forecasting wind speed and direction on the 1.3 km grid, selecting a legal site and layout for 55 IEA 22 MW turbines, and converting the wind forecast into farm-energy quantiles. Reproducibility code is available at `https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code`. The repository provides two Python entry points: `train.py` creates one artifact bundle from the permitted inputs, and `inference.py` loads that bundle to generate `predictions.csv` and `submission.json`. No previous submission is an input.

The forecast is a compact statistical correction of the supplied atmospheric predictors. Speed is estimated with lead-specific quantile models; direction is corrected in vector space to respect circular geometry; two component models map coarse wind to the target grid. A small set of context and external-trajectory corrections is applied only where historical replay supports it. The competition scorer matched all 4,196,640 submitted rows. The final interval scores were 9.511, 17.124, and 16.835 for speed and 76.400, 335.571, and 315.183 for direction at one, seven, and fourteen days, respectively. The rank-based primary score was 1.497276.

The farm centre is 54.109676°N, 0.952316°E. Its supplied-grid depth is 44.14 m. All 55 turbines satisfy the zone, depth, footprint, and spacing rules. Five-year replay with the prescribed PyWake configuration gives a mean capacity factor of 50.36%, mean annual energy production of 5,337.7 GWh, and mean wake loss of 7.10%. At the same centre, the selected layout improves mean capacity factor by 0.49 percentage points and reduces mean wake loss by 0.89 percentage points relative to a valid rectangular layout. The supplied cost model gives capital expenditure of EUR 4,478.4 million and mean-weather levelized cost of energy of EUR 88.69/MWh.

The report separates historical validation, definitive platform scores, deterministic legality checks, and financial scenarios. This separation matters: the definitive forecast score is observed, while hidden-climate farm yield and realized market profit are not. Section 7.3 states the resulting risks, their practical consequences, and the controls used to limit them.

## 1 Task, data, and evidence boundary

### 1.1 Required outputs

The forecast output contains lower, median, and upper estimates for wind speed and direction at one, seven, and fourteen days. The scored footprint has 43,715 points. With eight inference windows, three lead times, and four forecast hours, the required row count is

$$43{,}715 \times 8 \times 3 \times 4 = 4{,}196{,}640.$$

The siting output specifies one centre and 55 turbine coordinates. Each turbine must remain inside the 15 km by 15 km layout box, every pair must be at least 1,420 m apart, and the centre depth must not exceed 50 m. The turbine type and total capacity are fixed by the rules [1, 2].

### 1.2 Declared data use

Training uses the official 2016–2020 high-resolution target, low-resolution reanalysis, supplied European Centre for Medium-Range Weather Forecasts High Resolution Forecast tables, bathymetry, and static grids [3]. The definitive eight-window release is used only as an unlabeled inference input [4]. One external source is used for forecasting: causal GraphCast trajectories from the WeatherBench 2 archive. Here, causal means that each trajectory was initialized using information available at its issue time and contains no later observations. GraphCast is an ERA5-trained model and is sampled only from runs initialized at the relevant issue time [9, 10].

Historical target values are never available to `inference.py`. Reported evidence is separated into four categories:

- **Historical validation**: comparisons on held official target rows from 2016–2020.
- **Definitive forecast evaluation**: the six scores returned by the competition platform.
- **Deterministic audits**: row coverage, quantile ordering, siting legality, and energy bounds.
- **Scenario analysis**: economics and bidding calculations whose assumptions are stated explicitly.

### 1.3 Forecast score

Throughout the report, $\widehat{Q}_p$ denotes a conditional quantile at probability $p$. The speed interval is $[\widehat{Q}_{0.05}, \widehat{Q}_{0.95}]$, with median $\widehat{Q}_{0.50}$. For observation $y$, endpoints $l$ and $u$, and nominal error probability $\alpha = 0.10$, the linear interval score is

$$S_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}\{y < l\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}.$$

This proper score rewards narrow intervals only when they retain coverage [5, 6]. Direction uses the competition's circular counterpart because 0° and 360° represent the same direction. Lower scores are better. The primary score is rank based and is not the mean of the six raw scores.

## 2 Reproducible code and end-to-end execution

### 2.1 Code availability

The public repository at `https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code` contains `train.py`, `inference.py`, the pinned Python requirements, data-placement instructions, third-party licensing disclosures, this methodology report, and the final output archive. Competition datasets and external forecast fields are not redistributed. The README specifies where those inputs must be placed and gives the complete commands needed to regenerate the artifact and outputs.

The separation between training and inference is deliberate. Training may read labeled official rows from 2016–2020 and writes fitted parameters, validation decisions, the selected farm geometry, and the power-conversion policy. Inference reads only that artifact and unlabeled forecast inputs. It does not read historical targets, validation tables, or a previous `predictions.csv`. This makes the causal boundary inspectable in code rather than relying on a narrative claim.

### 2.2 Execution stages

The complete generation path is short enough to state directly.

1. `train.py` locates the official inputs and constructs causal features for each training issue time.
2. It fits the lead-specific speed quantiles, circular direction corrections, and coarse-to-fine vector mapping.
3. It replays optional corrections on held historical rows. Only retained policies and their activation rules are serialized.
4. The same training program screens legal farm centres, runs exact PyWake replay for a shortlist, evaluates layout candidates, and saves the selected geometry and power-conversion policy.
5. `inference.py` loads the artifact, predicts one window at a time, validates all forecast rows, derives 96 farm-energy triplets, and packages both required files.

| Stage | Inputs and computation | Persistent output and guard |
|---|---|---|
| Model fitting | Labeled official training rows, supplied predictors, and disclosed causal trajectory fields | Lead-specific probabilistic models; chronological and regime gates recorded in the artifact |
| Farm design | Official bathymetry, static grids, historical wind, and prescribed PyWake configuration | One legal centre, 55 coordinates, and deterministic constraint audit |
| Inference | Unlabeled official forecast windows and the saved artifact | Ordered wind quantiles and 96 farm-energy triplets; row and key checks must pass |
| Packaging | Generated forecast table and farm-design object | Archive containing exactly `predictions.csv` and `submission.json` |

**Table 1:** Execution contract. Labeled targets are confined to model fitting and validation; they are not inference inputs.

The production artifact contains 29 fitted estimators: nine lead-by-quantile speed models, six support-gated quantile refinements, five context endpoint models, four circular direction residual models, one shared spatial direction model, two conditional direction endpoint models, and two component downscalers. Every estimator is single-threaded. The number is reported to define the actual computational object, not as a performance claim.

The repository README records the exact commands and final parameter values. Reproduction requires one call to `train.py`, followed by one call to `inference.py`; neither command invokes a separate project script. The final configuration uses a six-day training sample, a 0.90 speed-coverage target, one-day speed-width scale 0.75, maximum seven-day centre-policy weight 0.4, one-day context-blend scale 1.25, and a 145° cap on the fourteen-day direction half-width. These values are serialized or passed explicitly, so the generated result does not depend on an undocumented interactive step.

# 3 Forecasting method

## 3.1 Speed and spatial downscaling

The core speed model is quantile model-output statistics implemented with LightGBM [7]. Separate models estimate $\widehat{Q}_{0.05}$, $\widehat{Q}_{0.50}$, and $\widehat{Q}_{0.95}$ at each lead time. Direct quantile fitting aligns training with the interval-score objective and avoids assuming symmetric or constant residual variance.

Predictors are limited to quantities with a physical or operational interpretation: supplied eastward and northward wind components, their magnitude and simple interactions, issue hour, seasonal sine–cosine terms, latitude, longitude, elevation, and coast distance. Wind components retain flow orientation; seasonal harmonics avoid an artificial discontinuity between the end and beginning of the year. Two additional regressors map coarse eastward and northward wind to the 1.3 km target grid. A residual calibration estimated from held historical data adjusts interval width. Finally, endpoints are sorted and clipped so that $0 \leq \widehat{Q}_{0.05} \leq \widehat{Q}_{0.50} \leq \widehat{Q}_{0.95}$.

## 3.2 Circular direction

Angular residuals cannot be fitted safely as ordinary numbers because a small error across north appears numerically close to 360°. The model therefore converts direction to sine and cosine components, learns residual corrections in that space, blends vectors, and converts the result back to degrees. This is standard directional-statistics practice [8]. Direction interval centres and widths are lead specific.

## 3.3 Different information at each lead

The available signal changes with forecast lead, so the three branches are not forced to share one correction.

- **One day**: supplied forecast fields provide the main signal. Lagged context corrects speed endpoints. Causal GraphCast wind at +24 to +42 hours supplies an additional direction trajectory.
- **Seven days**: supplied seven-day fields remain the anchor. A small context endpoint model and a gated external trajectory correction are active only on historically supported rows.
- **Fourteen days**: the supplied forecast does not extend to fourteen days. The branch combines the latest available forecast state with seasonal information and historically calibrated endpoint transformations. This is the least direct branch and is treated most conservatively.

The original contribution is not a new atmospheric simulator. It is the integration of lead-specific probabilistic calibration, circular correction, high-resolution vector mapping, and causal trajectory information under one reproducible selection and inference path. Its practical purpose is to improve the interval endpoints without building a large ensemble.

# 4 Validation and forecast results

## 4.1 Validation protocol

Candidate components are evaluated against the immediately preceding pipeline on exactly the same historical rows. The promotion rule has three levels: aggregate interval score must improve; every recorded held year must remain non-worse; and every populated spatial, speed, direction, interval-width, and signed-error regime must remain non-worse. Chronological years are kept separate wherever the archived audit permits. Unlabeled definitive inputs may be used only for a support check—for example, to reject activation patterns outside the historical range—and never for outcome-based selection. The definitive reference is never used for model selection.

Table 2 contains only comparisons for which the artifact retains absolute reference and candidate scores. Each row isolates one endpoint correction. The rows use different held periods and activation masks, so their changes are not added or compared as if they came from one common test set.

| Added component | Held data | Active rows | Reference | Candidate | Δ score |
|---|---|---|---|---|---|
| One-day speed context endpoints | 2019 | 61.87% | 9.975 | 9.645 | −0.330 |
| One-day speed context endpoints | 2020 | 74.97% | 13.040 | 11.468 | −1.572 |
| Seven-day speed context endpoints | 2016–2020 | 6.25% | 18.143 | 18.066 | −0.077 |
| Fourteen-day speed endpoints | 2019–2020 | 84.38% | 21.943 | 17.013 | −4.931 |

**Table 2:** Controlled historical ablations retained in the final artifact. Every row compares the same observations before and after one component; negative change is better. Rows are not additive because their held data and activation masks differ.

Coverage was retained for the two multi-year speed audits. Seven-day coverage changed from 94.20% to 94.20%; fourteen-day coverage changed from 94.58% to 91.75%. Both remain above the nominal 90% level, so the score reductions are not explained by simply narrowing the intervals below their stated coverage. The separate one-day

GraphCast direction replay retained only paired changes: circular interval score changed by $-2.329$ in 2018 and $-7.385$ in 2020. Because its absolute reference scores were not retained in the compact audit record, it is not mixed into Table 2.

Two examples clarify the selection rule. A stronger one-day external speed blend improved its mean score but failed localized issue-time checks and was rejected. A separate fine-grid speed-residual model failed its transfer gate and is not present in the production artifact. These rejected branches matter because they show that aggregate development score alone did not determine the final pipeline.

### 4.2 Interpretation of the fitted signal

Normalized LightGBM split gain is used as a descriptive audit of the fitted speed models, not as a causal explanation. At one day, the supplied forecast-speed variables account for 50.6% of normalized gain; location, vector components, and seasonal terms contribute 14.3%, 11.5%, and 11.3%. At fourteen days, direct forecast speed contributes only 5.4%, while seasonal terms, location, and wind-vector components together contribute 76.5%. This shift is consistent with the information boundary: direct forecast guidance is strongest at short lead, whereas the long-lead model relies more heavily on climatological and spatial structure.

Activation fractions provide a second, model-independent interpretation. The accepted one-day context correction modifies 61.87% and 74.97% of rows in the two retained folds; the seven-day correction is deliberately sparse at 6.25%; and the fourteen-day endpoint policy acts on 84.38% of its evaluated rows. These fractions explain why a small seven-day aggregate gain can still be credible and why the fourteen-day policy required stronger year and regime checks.

### 4.3 Definitive evaluation

The competition scorer matched 4,196,640 of 4,196,640 rows. Table 3 reports the complete definitive result.

| Interval score | One day | Seven days | Fourteen days |
|---|---|---|---|
| Wind speed | 9.511 | 17.124 | 16.835 |
| Wind direction | 76.400 | 335.571 | 315.183 |
| Platform primary score | | | 1.497276 |

**Table 3:** Definitive forecast scores returned by the competition platform. Lower values are better; the primary score is rank based.

The result is uneven. One-day direction is the strongest directional component, whereas seven- and fourteen-day direction remain the largest forecast weakness. The report does not infer calibration or generalization from the platform score alone.

## 5 Wind-farm siting and power conversion

### 5.1 Centre selection

The centre search first evaluates gross wind on the supplied grid because wake simulation at every cell would be unnecessarily expensive. Of 37,835 weather cells screened, 23,406 pass the geographic and depth rules. Fourteen candidates representing high mean wind, stable worst-year wind, balanced performance, and depth margin receive exact five-year PyWake replay. The final run retains the selected reference centre because no challenger improves mean and worst-year capacity factor while also satisfying the neighborhood, wake, depth, and economic gates. This is a robust selection rule, not proof of a global optimum.

### 5.2 Layout selection

At the selected centre, 16 valid layouts are replayed with the prescribed IEA 22 MW turbine and Bastankhah-Gaussian wake model [11]. The final geometry comes from constrained coordinate refinement and uses the available square while preserving spacing. Figure 2 shows the 55 submitted positions.

The legal audit is deterministic and is summarized in Table 4. The centre is inside the allowed zone and at 44.14 m depth. The layout uses almost all of the permitted north–south span, so coordinates are retained at full numerical precision in `submission.json`; rounding the outer coordinates would unnecessarily consume the remaining 0.06 m margin.

The selected centre is 69.07 km from the nearest coast in the supplied distance layer. Distance and depth enter the cost model through export-cable and foundation premiums. Economics is therefore used as a secondary discriminator between near-equal capacity-factor candidates; it does not displace a materially better yield candidate.

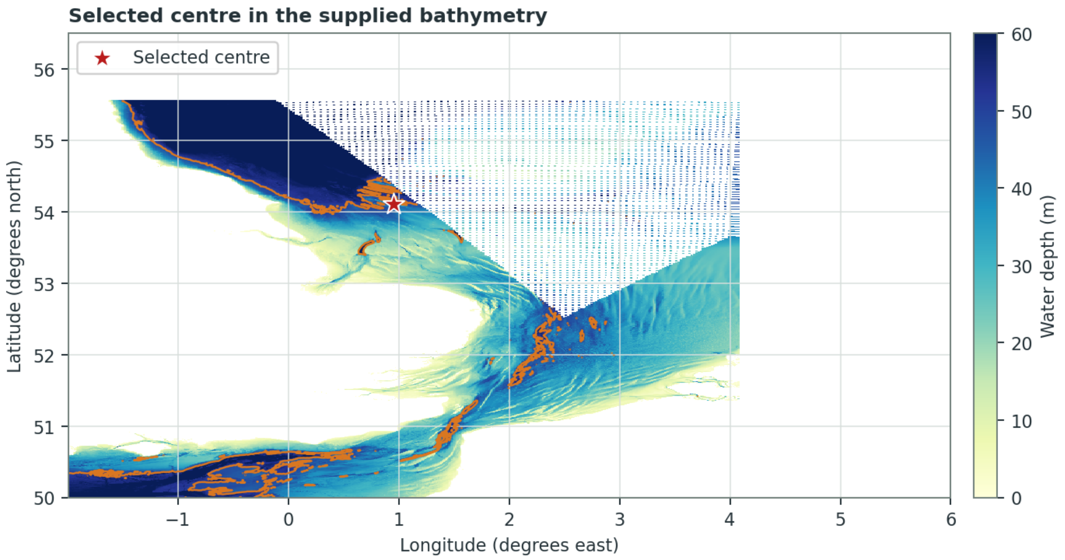


**Figure 1:** Selected centre on the supplied bathymetry. The contour marks the 50 m centre-depth limit.

| Constraint | Observed | Requirement | Margin |
|---|---|---|---|
| Turbine count | 55 | exactly 55 | 0 |
| Installed capacity | 1,210 MW | exactly 1,210 MW | 0 |
| Centre depth | 44.14 m | at most 50 m | 5.86 m |
| Maximum $\|x\|$ | 7,495.61 m | at most 7,500 m | 4.39 m |
| Maximum $\|y\|$ | 7,499.94 m | at most 7,500 m | 0.06 m |
| Minimum pairwise spacing | 1,554.64 m | at least 1,420 m | 134.64 m |

**Table 4:** Deterministic legality audit of the submitted centre and coordinates.

Table 5 isolates the value of the layout by holding the centre and five-year weather replay fixed. Relative to a valid rectangular 7-by-8 arrangement with one unused position, the submitted geometry adds 51.9 GWh/year on average and reduces wake loss by 0.89 percentage points.

| Layout at selected centre | Mean CF | Worst CF | Mean AEP (GWh/year) | Mean wake |
|---|---|---|---|---|
| Valid rectangular 7-by-8 | 49.87% | 47.73% | 5,285.8 | 7.99% |
| Submitted geometry | 50.36% | 48.18% | 5,337.7 | 7.10% |
| Change | +0.49 pp | +0.45 pp | +51.9 | −0.89 pp |

**Table 5:** Same-centre, same-weather comparison that isolates the submitted layout. CF is capacity factor; pp denotes percentage points.

The annual spread is material: capacity factor ranges from 48.18% to 54.95%, while wake loss remains between 6.93% and 7.31%. A directional stress test shifts the complete wind rose from $-45°$ to $+45°$; capacity factor remains between 50.23% and 50.38%, while maximum wake loss is 7.32%. This test shows that the irregular geometry is not narrowly tuned to one wind-rose orientation.

A second benchmark evaluates the combined centre-and-layout decision on public training data. A plain 7D grid at the public reference centre, snapped to 53.4982°N and 1.4928°E, gives 47.08% mean capacity factor, 4,989.9 GWh mean AEP, and 9.70% mean wake loss under the same five-year replay. The selected design gives 50.36%, 5,337.7 GWh, and 7.10%, respectively. This is explicitly a public-data proxy, not the organizer's hidden real-wind baseline; it is included because its inputs and simulation path are reproducible.

## 5.3 Farm-energy quantiles

For each inference time, `inference.py` extracts wind quantiles at the selected centre, applies the prescribed power-law shear from 125 m to the 170 m hub height, and runs the submitted geometry through the turbine power curve and wake model. The three paths are sorted after simulation and clipped to the physical six-hour range $[0, 7{,}260]$ MWh. The final JSON contains 96 values for each quantile. Their means are 738.9, 3,001.8, and 6,735.1 MWh for the lower, median, and upper paths.

This construction guarantees consistency with the submitted wind forecast and farm geometry. It does not guarantee calibrated farm-power coverage because a common median direction is used for all three speed paths and

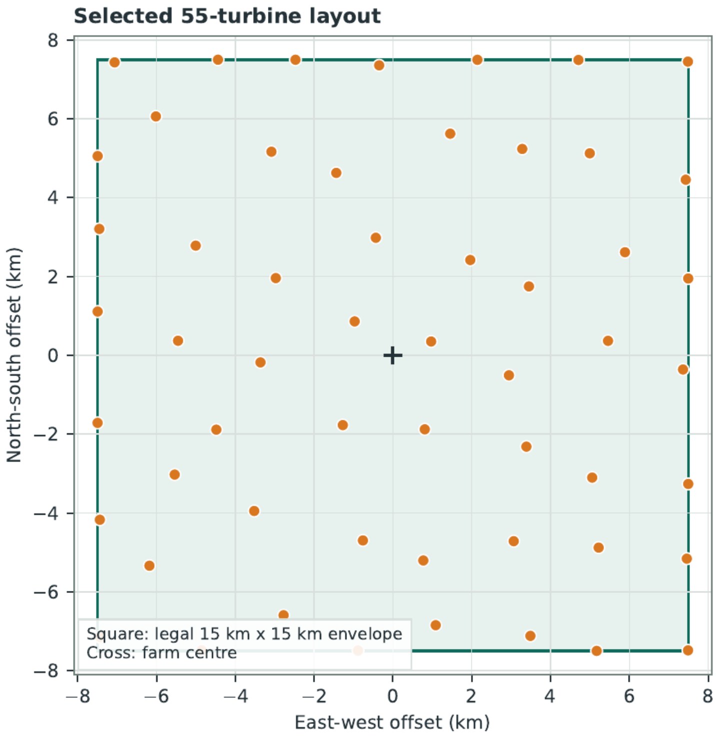


**Figure 2:** Submitted turbine coordinates relative to the farm centre. The dashed square is the permitted layout envelope.

| Replay year | AEP (GWh) | Capacity factor | Wake loss |
|---|---|---|---|
| 2016 | 5,106.9 | 48.18% | 7.31% |
| 2017 | 5,824.1 | 54.95% | 6.93% |
| 2018 | 5,279.1 | 49.80% | 7.00% |
| 2019 | 5,327.4 | 50.26% | 7.31% |
| 2020 | 5,150.7 | 48.59% | 6.97% |
| Mean | 5,337.7 | 50.36% | 7.10% |

**Table 6:** Exact annual PyWake replay of the selected site and layout using the supplied historical wind.

the joint speed–direction distribution is not modeled.

# 6 Economics and forecast value

## 6.1 Construction and operating cost

The supplied Phase 2 cost model is evaluated at the selected centre. Table 7 shows the complete decomposition used in the report. Depth and export distance are the site-dependent terms; no cost item is replaced with an external estimate.

| Capital component | EUR/kW | EUR million |
|---|---|---|
| Turbines and towers | 1,030.00 | 1,246.3 |
| Foundations, including depth premium | 812.10 | 982.6 |
| Array electrical system | 350.00 | 423.5 |
| Export cable, including distance premium | 552.56 | 668.6 |
| Installation | 350.00 | 423.5 |
| Development | 310.00 | 375.1 |
| Contingency | 296.47 | 358.7 |
| **Total capital expenditure** | **3,701.13** | **4,478.4** |

**Table 7:** Construction-cost result from the supplied Phase 2 cost model.

Annual operating expenditure is EUR 123.1 million. Using mean historical energy, the model gives a levelized cost of energy of EUR 88.69/MWh, comprising 65.63 EUR/MWh of capital recovery and 23.06 EUR/MWh of operation. Replacing mean energy with the lowest historical year raises the result to 92.70 EUR/MWh. This four-euro range is the direct financial consequence of the recorded interannual wind variability under the supplied assumptions.

### 6.2 Long-term price sensitivity

Revenue is not observed for the hidden siting climate, so profitability is presented as a scenario rather than a realized result. At mean annual energy of 5,337.7 GWh, constant energy prices of 70, 90, and 110 EUR/MWh imply gross annual revenue of EUR 373.6, 480.4, and 587.1 million. With capital expenditure $C_0$, annual energy $E$, annual operating cost $O$, energy price $P$, discount rate $r = 0.06$, and a 25-year operating life, the calculation is

$$\mathrm{NPV}(P) = -C_0 + \sum_{t=1}^{25} \frac{PE - O}{(1+r)^t}.$$

The corresponding pre-tax results are shown in Table 8.

| Energy price (EUR/MWh) | Gross annual revenue (EUR million) | Pre-tax NPV (EUR million) |
|---:|---:|---:|
| 70 | 373.6 | $-1{,}275.5$ |
| 90 | 480.4 | 89.2 |
| 110 | 587.1 | 1,453.9 |

**Table 8:** Transparent long-term price scenarios using the supplied cost result and mean historical energy. NPV excludes tax, degradation, curtailment, construction delay, and residual value.

The project is therefore close to break-even at the 90 EUR/MWh scenario and highly sensitive to price. This conclusion is more defensible than a single revenue forecast because it exposes the variable that dominates the decision.

The supplied operating-cost value is an aggregate allowance, not a component-level maintenance model. No turbine condition-monitoring, failure, vessel-availability, or repair-duration data are available in the permitted inputs, so a numerical predictive-maintenance saving would be invented. A deployable extension would use supervisory-control and condition-monitoring streams to estimate failure hazards and schedule vessel interventions; this report does not assign financial credit to that unavailable extension.

### 6.3 Quantile bidding

If shortfall and excess penalties are $\pi^+$ and $\pi^-$, respectively, the cost-minimizing bid is the predictive quantile [12]

$$\tau^* = \frac{\pi^-}{\pi^+ + \pi^-}.$$

For the supplied illustrative penalties of 90 EUR/MWh for shortfall and 30 EUR/MWh for excess, $\tau^* = 0.25$. The 25th-percentile energy bid is obtained by linear interpolation between the literal `predicted_q05` and `predicted_q50` arrays. Across the 32 one-day six-hour blocks, its mean is 2,487.1 MWh and the total commitment is 79.59 GWh.

This is a reproducible bidding rule, not a profit backtest. Realized farm production and imbalance settlement prices for the definitive blocks are unavailable, so assigning a realized financial gain would be unsupported.

## 7 Frugality, transparency, and residual risk

### 7.1 Compute footprint

Training and siting replay took 4.840 CPU-hours; inference and archive generation took 0.272 CPU-hours. No GPU was used. The measured machine used an Intel Core i7-7700HQ, and all estimators and numerical libraries were restricted to one thread. Table 9 converts wall-clock time using the processor's nominal 45 W thermal design power [13]. This is a CPU-package proxy, not whole-system electricity metering. Carbon uses a declared sensitivity of 0.45 $kgCO_2e$/kWh [14].

| Stage | CPU-hours | GPU-hours | kWh | $kgCO_2e$ |
|---|---:|---:|---:|---:|
| Training, validation, and siting | 4.840 | 0 | 0.218 | 0.098 |
| Inference and packaging | 0.272 | 0 | 0.012 | 0.006 |
| **Total** | **5.112** | **0** | **0.230** | **0.104** |

**Table 9:** Measured runtime and bounded CPU-package energy and carbon estimates.

The estimate excludes memory, storage, display, networking, cooling, and the upstream cost of producing the public GraphCast archive. Inference processes one window at a time and successful training deletes temporary checkpoints.

### 7.2 External resource disclosure

GraphCast is the only external forecast resource. The 37-level model is documented as trained on ERA5 from 1979 through 2017. The implementation is distributed under Apache-2.0 and the model weights under CC BY-NC-SA 4.0 [10]. Only the required causal forecast fields are sampled; model weights and forecast fields are not redistributed. No third-party operational forecast product is used beyond the official supplied forecasts.

### 7.3 Residual risk analysis

The useful distinction is between constraints that can be verified exactly and quantities that must transfer beyond the observed data. Row coverage, quantile ordering, turbine count, spacing, footprint, and centre depth are deterministic checks. Forecast skill on a new weather period, farm yield under the hidden wind record, and future project value remain uncertain. The following risks therefore qualify the corresponding claims rather than weakening unrelated parts of the method.

**Forecast transfer.** Component selection reuses a finite 2016–2020 record, so an apparently stable correction can still encounter a circulation regime absent from those years. Chronological fold, worst-year, and populated-regime gates reduce this risk; the support gate also prevents a correction from activating on definitive inputs outside its historical domain. These controls justify retaining the selected corrections, but they cannot establish calibration in an unseen regime. The definitive seven- and fourteen-day direction scores show that long-range direction remains the largest unresolved forecast error.

**Siting transfer.** The reported five-year PyWake replay uses supplied historical wind and gridded bathymetry, whereas the organizers evaluate yield with a hidden real wind record. Legality does transfer because it follows directly from the coordinates and supplied constraints. Yield does not transfer by construction. The centre is therefore selected using mean and worst-year capacity factor, neighbourhood stability, a 5.86 m depth margin, and directional-rotation stress tests rather than the best single-year AEP. These checks reduce dependence on one year or wind-rose orientation, but do not reveal unresolved local seabed or flow variation.

**Farm-energy uncertainty.** The submitted energy triplets propagate three speed quantiles through one median direction and the prescribed wake calculation. They are ordered and physically bounded, but they are marginal scenarios rather than samples from a learned joint speed–direction distribution. Consequently, they support a transparent quantile-bidding rule but not a claim of calibrated farm-power coverage. Joint probabilistic calibration would require matched realized farm power or an equivalent multivariate validation target.

**Economic scope.** The cost, LCOE, and NPV results answer different questions. The supplied deterministic cost model compares candidate locations; the historical energy replay measures technical variability; and the price table shows sensitivity to an explicit market assumption. Construction timing, degradation, curtailment, taxes, financing structure, failures, vessel access, and realized imbalance prices are not present in the permitted inputs. They are therefore excluded rather than assigned invented values. The financial results should be read as comparable scenarios, not as an investment-grade forecast.

The resulting priorities are specific. Forecast development should seek genuinely informative long-range directional fields without relaxing the transfer gates. Siting work should test the selected shallow-bank neighbourhood against additional allowed wind realizations while preserving the deterministic legality margins. Economic work becomes meaningful only when project-development and operating data are available at a resolution consistent with the decision being evaluated.

## 8 Conclusion

The method connects probabilistic wind prediction, legal wind-farm design, and farm-energy quantiles through a reproducible training and inference repository. Its strongest evidence is specific: complete definitive forecast coverage, controlled improvements for selected speed corrections, a legal 55-turbine design, a same-centre layout gain of 0.49 capacity-factor percentage points with lower wake loss, five-year yield and wake replay, and explicit cost and compute accounting. The code and generation instructions are available at https://github.com/MichaelIbrahim-GaTech/Sea-Wind-Phase-2-Code. Historical gates do not prove future calibration, and public replay does not prove hidden-climate yield. The main unresolved technical problem is long-range wind direction.

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